DATA-DRIVENSOLAR AEROPONICS

DESIGN, BUILD, AND IMPLEMENTATION PLAN

Project Lead: Jaime Conde and Shazia Khan
Team: Flora, Torrin, Citlaly, Luke

OBJECTIVE

Goal of this meeting:

1. Finalize required materials and identify missing components
2. Assign clear roles and responsibilities for each subsystem
3. Establish a realistic build timeline and work location
4. Confirm team commitment, availability, and long-term ownership

SYSTEM OVERVIEW

System Type

- Vertical aeroponic growing system
- Soil-free, mist-based root nourishment
- Sensor-driven (pH, TDS, temperature)

Flow System

- Water pumps
- Tubing + mist nozzles
- Flow meters

Structure

- PVC frame + vertical columns
- Reservoir tanks

Energy

- Solar panel (Optional)
- battery
- controller (Optional

Control System

- Raspberry Pi
- Sensors (pH, TDS, water level, light)
- Relay module + power supply

Lighting

- LED grow lights

Nutrients

- Nutrient solution
- Doser pumps
- Net cups + growing medium

RISKS AND MITIGATION

System Failure Risks

- Pump Failure (Flow System)
  → Loss of mist → plants die within ~30 minutes

Solution:

- Use reliable pump (rated continuous use)
- Keep backup pump available
- Test before full operation

- Clogged Tubing / Nozzles
  → Uneven or no nutrient delivery

Solution:

- Use filtered water
- Regular cleaning schedule
- Easy-access tubing design

- Power Failure
  → Entire system shuts down

Solution:

- Add battery backup (later phase)
- Manual check schedule
- Alert system (future upgrade)

Measurement & Control Risks

- Sensor Failure (pH / TDS / Water Level)
  → Incorrect nutrient conditions

Solution:

- Manual verification (weekly)
- Calibrate sensors regularly
- Redundant checks (visual + measurement)

- Relay / Electrical Failure
  → Pump or system not triggered

Solution:

- Test wiring before deployment
- Use fused connections
- Keep spare relay module

Project Execution Risks

- No Summer Maintenance
  → System dries out and fails

Solution:

- Assign weekly maintenance roles
- Create schedule before build
- Transfer ownership if needed

- Delayed Structure Build
  → Entire project timeline pushed back

Solution:

- Prioritize structure materials first
- Assign structure lead
- Set early build deadline

BOM

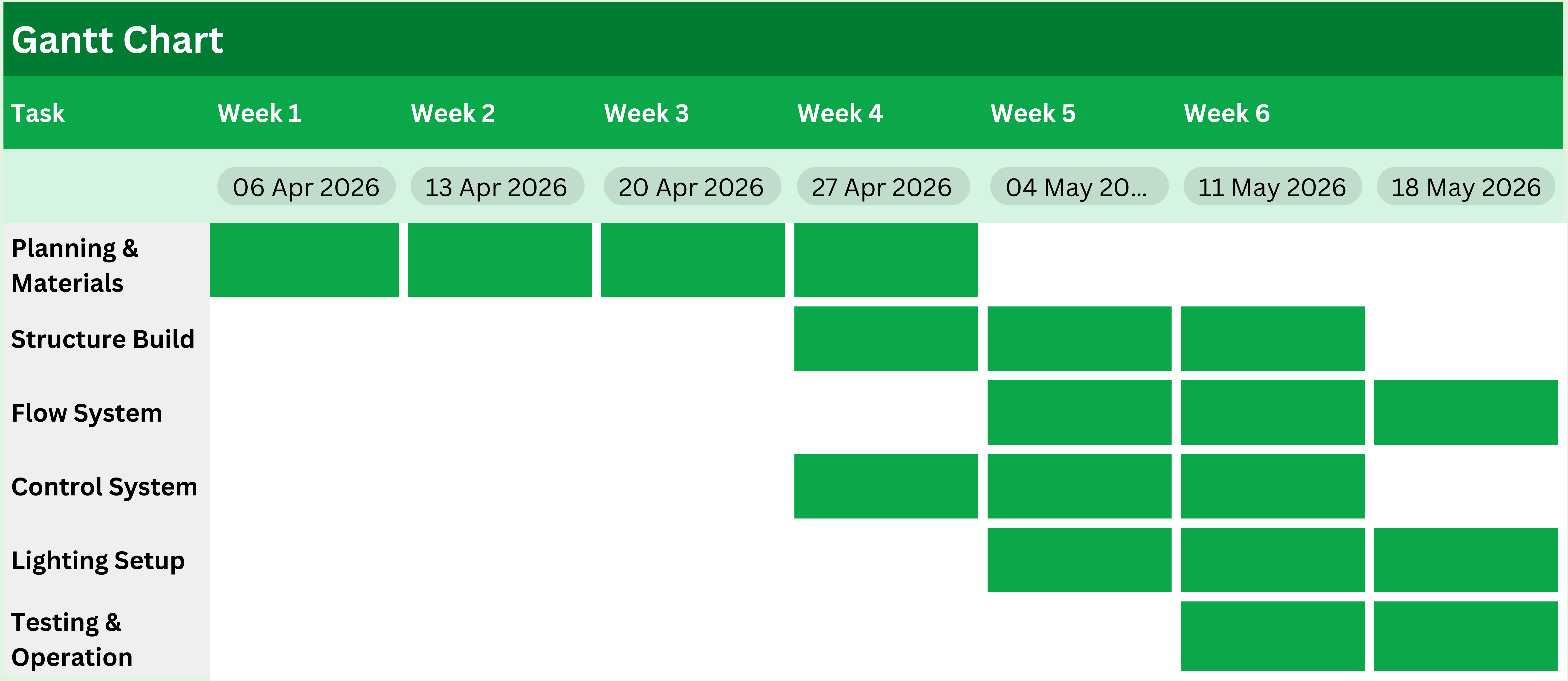
Category	Component	Typical Spec Assumed	Qty	Priority	Have?	Price ($/unit)	Link	Used Option?	Notes	Bought, Status? (Y/N) (Arrived, Arriving)
Structure	Base			HIGH	X		https://a.co/d/0fSfPl			
Structure	Food grade PETG	For main plant base hub	3	HIGH		20	Food-Grade PETG	NO	Jamie will 3-D print plant base hub	Arrived
Structure	Reservoir tank (15–25 gal)	Food-grade plastic	2	HIGH		69	https://a.co/d/05Tz	YES		
Flow	Water pump (12V DC, 800–1200 L/h)	Submersible or inline	1	HIGH		30	Pump	YES	critical	
Flow	Flow meter (Hall-effect 1/2")	YF-S201 style	2	MED	X	15		YES		
Flow	Spray Nozzles	Hydroponic tubing + atomizers		HIGH		10	Spray Nozzles			
Sensors	Water level sensor	Float switch / ultrasonic	2	MED	X	10		YES		
Sensors	Light intensity sensor	BH1750 / TSL2561	1	LOW	X	15		YES		
Sensors	pH sensor	Probe + signal board	2	HIGH	X	55		YES		
Sensors	TDS sensor	Analog probe + board	1	HIGH	X	35		YES		
Sensors	Time interval (1min)		1	HIGH		12	Timer			
Electronics	Raspberry Pi (4 or 5)	4GB version	1	LOW	X	80		YES		
Electronics	Relay module	4–8 channel 5V relay		LOW						
Electronics	DC-DC converter (12V?5V,5A)	Buck converter	1	LOW	X	20		YES		
Electronics	Power supply	12V regulated	1	HIGH						
Lighting	LED hanging light	Full spectrum hydroponic	2	HIGH	X	50	https://a.co/d/02Tpc BKVhttps:/a.co/d/04	YES		
Energy	Solar panel (100W)	Monocrystalline	2	LOW		50		YES	optional early	
Energy	Battery (12V AGM 35–55Ah)	Deep cycle	1	LOW		80		YES		
Energy	Solar charge controller	10–20A	1	LOW	X	30		YES		
Nutrients	Nutrient Solution	Hydroponic nutrient mix		MED		16	Nutrient Solution		Maxi Grow	
Nutrients	pH up and down kit	pH regulator	1	MED		19	pH Regulator			
Misc	Doser pump (peristaltic, 12V)	40–100 mL/min	x/8	MED	X	15		YES		
Misc	Net cups	2"–3" size		HIGH			Net Cups			Encase they dont work:
Misc	Growing medium	Clay pebbles / coco		HIGH			Included with Net Cups			
					Total:	$646	Total W/ Supplies:	$301		

TOOLS AND SPONSORSHIPS

	Tool or Supply	Who Has It	Need to Buy?	Link	Notes
Drill	Plug in cord type	Flora, Jaime Conde			
Hole saw (net cup size)	Circle cutter for the drill **depends** diameter needed	Flora			
PVC cutter					
Soldering kit		Jaime Conde			
Multimeter	kobalt Analog	Flora, Jaime Conde			
Wire stripper		Flora, Jaime Conde			
Zip ties / clamps	Some… **depends** on length needed	Flora, Jaime Conde			
Silicone sealant		Jaime Conde			
Metal Hinges	Supply	Shazia	Yes		

Store Name/People	Location	Contact	What They Sell/Give	Sponsorship Potential	Notes
Corrine	Her garage	(Shazia will communicate via text)	PVC?	Yes	
Atlantis Hydroponic	1409 Howell Mill Rd NW, Atlanta, GA 30318	(404) 367-0052 atlantishydroponics.com		Maybe	Need to contact
Dr. Rothschild	Near Dunwoody Campus		POTENTIALLY (still need to confirm) could be a build location	Maybe	

BUILD PROCESS



ROLES AND RESPONSIBILITIES

Role	Primary Responsibility	Key Tasks	Priority	Member Responsible
Structure Lead	Build physical system base	PVC frame, columns, reservoir setup	High	**Jaime Conde, Flora, Citlaly**
Flow System Lead	Ensure water delivery (CRITICAL)	Pump, tubing, mist nozzles, flow meter, leak testing	Critical	**Jaime Conde, Flora**
Control / Electrical Lead	Basic system wiring	Relay, power supply, sensors (optional phase)	Medium	**Citlaly M, Jaime Conde, Lucas Lombardi**
Lighting Lead	Install plant lighting	LED grow lights, positioning, coverage	High	**Jaime Conde, Flora, Citlaly**
Procurement Lead	Materials sourcing & tracking	Find parts, compare prices, manage purchases	Medium	**Shazia Khan**
Maintenance Lead	System upkeep & plant care	Water levels, nutrients, net cups, growing medium	High	**Jaime Conde, Shazia Khan**

THIS CHART IS MEANT AS A GENERAL REPRESENTATION OF WHAT TASKS NEED TO BE DONE, NOT A STRICT ASSIGNMENT SHEET. I ENCOURAGE PEOPLE TO WORK TOGETHER IN AN INTERCONNECTED WAY. IF WE KNOW SOMEONE WILL BE GONE FOR A SPECIFIC WEEK/TIME PERIOD WE CAN MOVE AROUND WHO DOES WHAT.